

LATHER HIGH SCHOOL

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QA Testing Verification Report

Phase 2 Functionality, API Integration, and Device Optimization Checks

DOCUMENT TYPE

TECHNICAL QA AUDIT

PREPARED BY

Royal Crown Consultancy Services & @RCCS
Global

PREPARED FOR

Lather High School

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1. Quality Assurance Overview

To ensure reliability, scalability, and code correctness, a systematic testing audit was conducted across all newly added and modified features. The verification process covered API routing constraints, input validation criteria, responsiveness across device break-points, and production build checks.

2. Backend API Endpoint Verification

The backend API routes were tested using HTTP mock payloads to ensure secure data transfer and logging:

Route / Endpoint	Method	Test Parameter	Expected / Actual Result
/api/cms/admission	POST	Submit complete student details	HTTP 200 OK. Admission request logged successfully.
/api/cms/admission	POST	Submit empty required name/phone	HTTP 400 Bad Request. Missing parameters error returned.
/api/cms/faculty	GET	Fetch all teaching staff records	HTTP 200 OK. Returns JSON array of seeded faculty.

3. Form Validation & Client State Machine

The online admission form at `/admission` was tested against user interaction states:

- **Required Inputs Test:** When attempting to submit without entering Student Name, Parent Name, Gmail, or Contact Number, the form execution halts and renders a red alert box: *'Please fill in all required fields.'*
- **Loading Spinner Test:** During form dispatch, the submit button changes text to *'Submitting...'* and disables itself to prevent duplicate API posts.
- **Success Receipt Rendering:** Upon successful backend submission, a confirmation card is displayed, showing a summary of the registered data and a checkmark animation.

4. Responsive Design & Layout Verification

Each modified component was tested across various device screen resolutions to guarantee visual elegance:

- **Desktop Displays (> 1024px):** Profile cards align in 3-column grids, and footer columns align side-by-side with appropriate spacings.
- **Tablet Screens (768px - 1024px):** Staff grid seamlessly collapses to 2 columns. Footer scales to 2x2 grid.

- **Mobile Handsets (< 768px):** Forms and profile grids stack into a single column. The header menu hides behind a touch-activated sidebar drawer containing all updated links.
- **FAQ Accordions:** Transition animations tested on mobile viewport dimensions to verify text wraps without horizontal scrolling.

5. Production Build & Compiler Validation

The system code was subjected to strict TypeScript verification and optimization builds:

- **Backend Compilation:** Compiling code using the tsc compiler completed with zero syntax warnings or route configuration errors.
- **Frontend Compilation:** Executing Next.js production compiler (next build) successfully optimized and outputted static HTML assets for /admission, /faq, and /staff.